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Dimensionality reduction and foundation-model embeddings

Week 7: The hinge of the module: from features you chose to features a model learned.

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By the end of this week you should be able to

Learning outcomes served: LO1, LO2, LO4

- Explain the curse of dimensionality and its practical consequences for biological data.
- Perform and interpret PCA, including the explained-variance spectrum and loadings.
- Distinguish PCA from neighbour-embedding methods such as t-SNE and UMAP, and state what each preserves.
- Explain what a foundation-model embedding is and how it is produced.
- Compare a hand-engineered representation with a learned one on the same data, and identify which task favours which.

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The biological problem

Part 1 · Why anyone cares about this week's method

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Biological context

- Biological data is routinely wider than it is tall. A transcriptome has 20,000 genes measured in perhaps 200 patients; a protein is a sequence of hundreds of residues drawn from an alphabet of twenty. Both need to be reduced to something a model can work with.

- The classical answer is to project onto directions of maximal variance. The modern answer is to let a model that has read a great deal of biological sequence tell you where the sequence sits in a learned space. This week puts the two side by side.

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The method

Part 2 · Concepts, then the mathematics

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The curse of dimensionality

As dimension grows, distances between points concentrate and the volume of space grows faster than any feasible sample. Nearest neighbours stop being near.

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Principal component analysis

An orthogonal linear projection onto the directions of greatest variance. It is deterministic, invertible, and its components are interpretable through their loadings. On WDBC, the first two components carry 63.2 per cent of the variance and the first six carry 88.8 per cent.

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Key concepts

Explained variance How much of the total variance each component captures. A slowly decaying spectrum means the data genuinely is high-dimensional and no small projection will be faithful.

t-SNE and UMAP Non-linear neighbour-embedding methods that preserve local neighbourhoods.

Representation learning Instead of choosing features, learn a mapping from raw input to a vector space in which the geometry is useful.

Embedding The vector a model produces for an input. For a protein language model, mean-pooling the final hidden layer over residues gives one fixed-length vector per protein, whatever its length.

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Key concepts (continued)

Self-supervised pretraining Training by predicting masked parts of the input. No labels, so the corpus can be enormous: DNABERT-2 used 32.49 billion bases from 135 species.

Anisotropy Language-model embedding spaces are dominated by a few directions. This affects nearest-neighbour search and often makes centring or whitening worthwhile before using the embeddings downstream.

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Principal component analysis

$$C = (1/(n-1))\, X^T X$$

$$C\, v_k = \lambda_k\, v_k$$

Standardise, form the covariance matrix, and take its eigenvectors. The k-th principal component is the direction of the k-th largest variance, orthogonal to all the previous ones.

- X the centred, standardised data matrix, n × p
- v_k the k-th eigenvector: the loadings of component k
- λ_k the k-th eigenvalue: the variance along that direction
- **explained** $\lambda_k / \Sigma\lambda$ is the proportion of variance captured

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Producing a per-sequence embedding

$$e = (1/L) \cdot \Sigma\ h_i \qquad \text{for real residue positions } i$$

Mean-pool the final hidden layer over the residues, excluding the special tokens. One fixed-length vector per protein, whatever its length, and computed without using any of your labels.

- h_i the final-layer hidden state at position i
- L the number of real residues, excluding BOS, EOS and padding
- e 320 dimensions for ESM-2 8M, 1,280 for the 650M model
- **note** your labels are never used, so this is a fixed transformation, not a fitted step

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PCA on the real WDBC data, measured

The PCA is refitted inside each cross-validation fold, which is the only correct way to measure this.

Components	Cumulative variance	CV accuracy
1	44.3%	92.6%
2	63.2%	94.3%
5	84.7%	96.5%
10	95.2%	97.7%
20	99.5%	97.5%
30 (all)	100%	97.4%

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The two tasks, and why they disagree

Identical classifiers throughout, so any difference is attributable to the representation. The family task is defined by composition; the triad task is defined by order, and both classes have identical composition by construction.

Task	Engineered features	Learned embedding
Protein family (4 classes, chance 25%)	100.0%	97.5%
Ordered H-D-S triad (binary, chance 50%)	51.9%	89.7%

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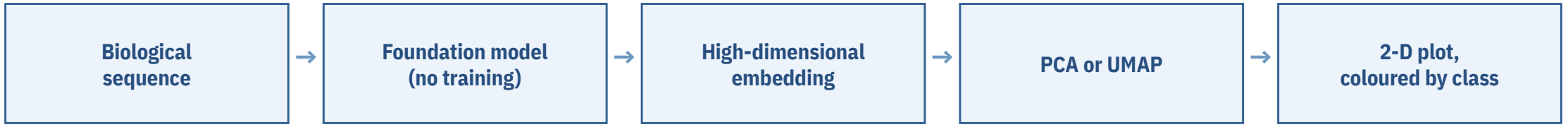
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From sequence to a two-dimensional picture

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You look at two dimensions. You should model in all of them.

The pipeline



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Reading a projection honestly

The most common misuse in the literature is reading a gap between two UMAP clusters as a measure of biological difference.

- PCA is linear, deterministic and invertible; its distances mean something and its loadings are interpretable
- t-SNE and UMAP preserve local neighbourhoods and destroy global geometry
- In a neighbour embedding, between-cluster distance, cluster size and empty space are all uninterpretable
- Language-model embedding spaces are strongly anisotropic, dominated by a few directions
- Centring or whitening before a distance-based method is usually worth it
- The plot is a diagnostic, not a result: if classes separate, that is a hypothesis to test with a classifier and a held-out set

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Worked example

Part 3 · The pipeline, stated concretely

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The pipeline, stated concretely

- Biological sequence, for instance a 260-residue protein.
- Into a foundation model: tokenise (single residues for ESM-2, byte-pair-encoded fragments for DNABERT-2, six-mers for the Nucleotide Transformer) and run a forward pass. No training happens, and your labels are never used.
- Out comes a high-dimensional embedding: one vector per token, mean-pooled over real residue positions into a single vector per sequence. ESM-2 at 8 million parameters gives 320 dimensions; the 650-million-parameter model gives more.
- Then PCA or UMAP down to two dimensions, purely so a human can look at it.

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The pipeline, stated concretely (continued)

- Then the plot, coloured by biological class. And here is the discipline: the plot is a diagnostic, not a result. If classes separate, that is a hypothesis worth testing with a classifier and a held-out set.
- If they do not, the signal may still be there in the full-dimensional space, since a two-dimensional projection of a 320-dimensional space discards a great deal.

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You look at two dimensions. You should model in all of them.

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Try this interactively

PCA Visualiser

How much is lost by reducing, and does the loss matter downstream?

`weeks/week07.html` · `Interactive exploration`

Open the week page on the course website and scroll to Interactive exploration.

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Try this interactively

Foundation Model Embedding Explorer

When does a learned representation beat hand-engineered features, and when does it not?

`weeks/week07.html` · `Interactive exploration`

Open the week page on the course website and scroll to Interactive exploration.

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The foundation-model connection

Part 4 · Two tasks, two answers

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Two tasks, two answers

- The embedding explorer on this page carries two different labels for the same sequences, and they behave in opposite ways.
- The family label is driven by amino-acid usage. A twenty-dimensional composition vector describes it almost perfectly, and hand-engineered features match or beat the embedding. The right answer here is to use the simple thing.
- The ordered-triad label is driven by residue order alone: both classes contain exactly the same residues at the same positions, differing only in their N-to-C arrangement. Composition is identical by construction, so any bag-of-residues method must sit at chance.
- Only a representation that encodes order can do better.
- That is the whole argument for learned representations, in one dataset. Not "the embedding is better", but "the embedding can see something composition cannot, and whether that matters depends entirely on your task".

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Interpretation and practical

Part 5 · What the result means, and what you will do with it

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Your UMAP of patient embeddings shows two well-separated blobs. Before you write that you have found two disease subtypes, list three non-biological explanations that would produce exactly the same picture, and say how you would rule each out.

Think like a scientist. Bring your answer to the laboratory.

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PCA and the geometry of a learned representation

This week's laboratory · Reduce a real dataset properly, then compare an engineered representation with a learned one on identical footing. · 2 hours

- Standardise WDBC and run PCA. Plot the explained-variance spectrum and the cumulative curve. How many components are needed for 90 per cent?
- Plot PC1 against PC2 coloured by diagnosis. Then inspect the PC1 loadings: which original features dominate it, and what do they have in common biologically?
- Fit logistic regression on the raw 30 features and on 2, 5, 10 and 20 principal components, all inside a Pipeline. Plot accuracy against the number of components. Where is the knee?
- Load the protein teaching set from js/data/sequences.js. Note the simulation warning and record in your notebook what is simulated and what is not.
- For the family label, run k-NN on the engineered representation and on the embedding. For the hasSite label, do the same. Produce one table with four numbers.
- Explain the pattern in that table. Which task favours which representation, and what property of each task explains it?

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Self-check

- On the WDBC data, the first two principal components explain 63.2 per cent of the variance. What does the remaining 36.8 per cent tell you?
- A) The PCA failed
- B) A two-dimensional plot is a substantial simplification and structure may be present in the discarded directions
- C) The data should be scaled differently
- D) Two components are sufficient

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Self-check

- Which statement about a UMAP plot is correct?
- A) Distances between clusters reflect true dissimilarity
- B) Cluster sizes reflect the number of points
- C) Local neighbourhoods are approximately preserved, but between-cluster distances and empty space are not interpretable
- D) It is a linear projection

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References and further reading

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Questions

Next week: continue to Week 8.

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